

2(a)(i)	any one from: idea of compare / AW, observed and expected (data / results / ratios / phenotypes) ; or data is, categoric / discrete / discontinuous / nominal ;				1																								
2(a)(ii)	idea of no (significant) difference between the observed and the expected (results) ; or idea of difference between observed and expected (results) is due to chance ;				1																								
2(a)(iii)	<table><tr><td>offspring phenotype</td><td>O</td><td>E</td><td>$\frac{(O-E)^2}{E}$</td></tr><tr><td>wrinkled wings rosy eyes</td><td>44</td><td>36 (36.000)</td><td>1.78 (1.77778)</td></tr><tr><td>wrinkled wings red eyes</td><td>2</td><td>12 (12.000)</td><td>8.33 (8.33333)</td></tr><tr><td>normal wings rosy eyes</td><td>2</td><td>12 (12.000)</td><td>8.33 (8.33333)</td></tr><tr><td>normal wings red eyes</td><td>16</td><td>4 ; (4.000)</td><td>36.00 ; (36.00000)</td></tr><tr><td colspan="3">$\chi^2 =$</td><td>54.44 ; (54.44444)</td></tr></table> apply ecf as needed				offspring phenotype	O	E	$\frac{(O-E)^2}{E}$	wrinkled wings rosy eyes	44	36 (36.000)	1.78 (1.77778)	wrinkled wings red eyes	2	12 (12.000)	8.33 (8.33333)	normal wings rosy eyes	2	12 (12.000)	8.33 (8.33333)	normal wings red eyes	16	4 ; (4.000)	36.00 ; (36.00000)	$\chi^2 =$			54.44 ; (54.44444)	3
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2(a)(iv)	n–1 <u>and</u> probability of 0.05 / one less degree of freedom than number of categories <u>and</u> probability of 0.05 ;				1																								

Question	Answer	Marks
2(a)(v)	<i>apply ecf as needed</i> 1 reject (null hypothesis) and state that, the calculated value / χ^2, is higher than, 7.82 / the critical value ; 2 <i>If null hypothesis rejected:</i> (between O and E): there is a significant difference / the difference is not due to chance <i>idea that</i> there is a less than 5 % probability that the difference is due to chance / there is a more than 95% probability that the difference is not due to chance ; 3 <i>From mp1 If null hypothesis rejected or mp2 If there is a significant difference / AW, stated</i> <i>idea that:</i> genes, are on the same chromosome / are not on different chromosomes / are linked / show autosomal linkage / do not assort independently ; A <i>idea of, 'may be / can be', throughout mp3</i>	3